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Report 2:

Literature Review

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CASSIA: A Constraint-Based Intelligent Agent for Coordinating Solar, Waste-Heat, Cooling, and Water Sourcing in AI Data Centers

1. Introduction

In 2024, all data centers worldwide used about 415 TWh of electricity, about 1.5% of the world total. In its Base Case, the IEA projects about 945 TWh by 2030, with AI as the largest driver [1]. Li et al. project that global AI may withdraw 4.2–6.6 billion m³ of water in 2027; withdrawal is water taken from sources, and only part of it is consumed. They also show that saving carbon does not always save water [2]. Water is also poorly measured: Mytton reports that fewer than a third of data center operators track their water use [3].

Our project, CASSIA, is an agent that plans every hour. It shares solar power and waste heat between AI jobs, cooling, and alternative water production by membrane distillation (MD) or forward osmosis (FO). The goal is less grid power and city water, with all jobs on time, servers within safe temperatures, and zero thermal-throttling events for high-priority jobs.

We searched IEEE Xplore, the ACM Digital Library, ScienceDirect, and Google Scholar (2021–2026), combining "data center" or "scheduling" with terms such as "water," "cooling," "solar," "waste heat," "constraint programming," "A*," "MDP," "fuzzy," "Bayesian network," and "model predictive control." We kept peer-reviewed papers with a tested method. We review 21 studies [4]–[23], [25], grouped by method, and two integrated works [27], [28] in Section 4. Sources [1]–[3], [24], and [26] give background. Two studies come from other fields ([6] factories, [13] EV charging); we reuse only their problem structure. Most studies use classical AI or optimization at their core; [15] and [27] use machine learning (ML) and are comparators only, since CASSIA's core is classical AI.

2. Related Work

2.1 Scheduling jobs around clean power

Schweisgut et al. [4] schedule tasks on a cluster with a changing green power budget, testing 34 workflows of up to 30,000 tasks. With a deadline, the problem is easy on one processor (dynamic programming) but NP-hard on many. So they use greedy rules plus hill-climbing local search, and compare with an exact ILP only up to about 200 tasks, because the solver becomes too slow.

Jiang et al. [5] built WaterWise. It places batch jobs in data centers in different regions using mixed-integer linear programming (MILP) that weighs carbon and water. Delay limits are soft: if no plan meets them, it pays a penalty. Terbrack and Claus [6] use constraint programming (CP) to schedule factory jobs under changing electricity prices. Due dates and setup times are constraints. On 20 test cases, energy cost fell by 5.3%, peak demand by 11.8%, and emissions by 8.3% on average. The goals conflict, so a person must rank them before solving.

Wiesner et al. [7] study "temporal shifting": delaying flexible jobs to hours with cleaner power. They analyze grid carbon data from Germany, Great Britain, France, and California for 2020, and simulate shifting strategies under different time limits and forecast errors. The benefit depends strongly on the region and on how long a job may wait. Radovanović et al. [8] describe Google's deployed system. Each day, a risk-aware optimization sets hourly capacity limits for flexible work in every cluster, using carbon and demand forecasts, while all flexible work still finishes within the day. In one reported example, average cluster power fell by 1–2% in the highest-carbon hours.

Other studies add on-site solar and storage. Haghshenas et al. [9] schedule mixed workloads while accounting for solar power, the cooling system (including free cooling), and the electricity tariff; energy cost fell by up to 46% against earlier methods in trace-based simulation. Peng et al. [10] use a rule-style scheduler, REDUX3: it fills spare solar power with waiting jobs, delays non-urgent jobs when solar is low, and charges or discharges UPS batteries by price. Da Silva et al. [11] use MILP to move both jobs and stored energy between solar-powered mini data centers; grid electricity fell by up to 22% against moving jobs alone. Haddad et al. [12] use MILP to plan solar, wind, batteries, and hydrogen for a data center with no grid connection, over 72-hour windows with weather forecasts.

None of these studies controls cooling modes or produces water. [9] models cooling power but does not reuse heat, and [5] models water but cannot change how the building is cooled.

2.2 Heuristic search and thermal-aware placement

Shohan et al. [13] plan electric vehicle charging in buildings with solar power using A* search. The cost so far, g, is the energy cost at the real-time price, and the estimate h is forecast need times forecast price. The authors call h admissible but do not prove it; with solar and vehicle-to-grid sales, the remaining cost can fall below need × price, so optimality is unproven.

Rostami et al. [14] place work on servers so that the room needs less cooling. They make the heat and power models linear, write the problem as an integer program, and add a heuristic because the exact solver is slow. They validate it on a detailed model of a real data center; letting idle servers run hotter than busy ones cut cooling power by more than 30% against a simpler baseline. Qi et al. [15] built SHIELD for 16 geographically distributed data centers. It combines local search with an evolutionary algorithm, and an ML model picks where the local search starts. It returns trade-off plans for energy cost, carbon, and water footprint.

2.3 Rules, MDPs, Bayesian networks, and predictive control for facilities

Borkowski and Piłat [16] built a rule-based controller that decides when chillers switch between free cooling and compressor cooling. Its rules come from past plant data and were validated on an operating cooling system through 2017. Mohsenian et al. [17] use two fuzzy rule-based controllers, one for air dampers and one for cooling-unit fan speed, in a real test data center (a small neural network models the dampers). Against the worst case, they saved up to 75% of fan power and 16% of chiller power in one scenario.

Bayati [18] models server power control as a Markov decision process (MDP). Value iteration finds the best policy, tested with Google job traces; the largest case has 103 servers. Wang et al. [19] use a hybrid Bayesian network to diagnose chiller faults. It combines model residuals, expert knowledge, and sensor data, and still works if one type of evidence is missing; on two test chillers it beat common methods.

Model predictive control (MPC) re-plans over a moving time window. Zhu et al. [20] use MPC with MILP to run chillers and a cold-water storage tank under free, hybrid, and mechanical cooling modes. In a Guangzhou case at 55% IT load, yearly energy fell by 12.19%, 4.04%, and 22.15% in the three modes against a baseline. Zhu et al. [21] use stochastic MPC for a data center microgrid: possible solar, wind, and load values become scenarios, batch jobs are scheduled a day ahead, and the plan is corrected during the day. Liu et al. [22] model three kinds of delay-tolerant jobs and schedule them a day ahead to cut operating cost and wasted renewable power.

2.4 Using waste heat for water and heat

Yuan et al. [24] review data center waste heat recovery for heating, district heating, cooling, and power. Hou et al. [25] use MPC to run a university heating system fed by data center waste heat and a water storage tank. Against rule-based control, monthly energy cost fell by up to 3.2%, and the data center's chilled water stayed more stable.

Yang et al. [23] co-plan a data center and a reverse osmosis (RO) plant less than 1 km apart in Jubail, Saudi Arabia. They schedule data center tasks, and waste heat warms the seawater, which changes RO efficiency. They solve a two-stage stochastic MILP by Benders decomposition. Total yearly cost fell 86.5% against conventional designs, 17% of it from heat reuse. But the heat only preheats RO, which runs on electricity; it does not drive MD, FO, or storage, and its summary reports no cooling-mode choice. Giagnorio et al. [26] compare FO combined with RO and FO combined with MD, and find FO-MD more versatile, partly because MD can use low-grade heat. We found no peer-reviewed study that links data center waste heat to FO.

3. Critical Comparison

Table 1 summarizes the studies. Datasets and baselines differ, so percentages show direction only. Only [4] checks against an exact optimum, and only [8], [16], and [17] report tests on real systems; the rest are simulations or planning studies.

Optimality vs. speed. Exact methods are optimal for their model but grow fast: if n jobs each pick one of T start slots, there are up to T^n schedules. So [4] uses its ILP only up to about 200 tasks, [14] adds a heuristic to its integer program, and the MDP in [18] grows with servers × queue × lost jobs. MPC [20], [21], [25] keeps problems small by planning a short window and re-planning, which is the same idea as CASSIA's hourly re-planning.

Hard vs. relaxed constraints. CP [6] keeps due dates hard but ranks its goals in a fixed order chosen beforehand, so a slightly costlier plan with a much lower peak is never chosen. WaterWise [5] relaxes delay limits with a penalty, which keeps a plan feasible but allows late jobs. Google's system [8] keeps a hard daily rule: all flexible work finishes within the day.

Deterministic vs. uncertain. Uncertainty is handled in different ways: scenarios in [21], a two-stage stochastic model in [23], risk-aware limits in [8], and probabilities in the MDP [18] and the Bayesian network [19]. [7] tests how forecast accuracy affects shifting. Most other studies plan on point forecasts, and none tests solar forecast errors together with cooling and water decisions.

Inspectable vs. learned. In rules [10], [16], fuzzy rules [17], CP [6], MILP [5], [11], A* [13], and Bayesian networks [19], each decision can be traced to a rule, constraint, cost, or probability. SHIELD [15] reached the next-best method's plan quality with 34.4× less computing time, but learned guidance is harder to trace.

Table 1. Summary of the reviewed studies and the two closest integrated works.

Ref	AI Technique	Problem / Test Cases	Evaluation & Result	Limitation
[4]	Greedy + local search; ILP check	Carbon-aware scheduling with a deadline; 34 workflows, up to 30,000 tasks	Median about 0.6 of ASAP baseline; about 0.15 for one heuristic at 3× deadline	Fixed task mapping; heuristics not optimal
[5]	MILP weighing carbon and water; soft delay limits	Job placement across regions; Google and Alibaba traces	About 21% less carbon, 14% less water vs. an unaware scheduler	Weak baseline; cooling not controlled
[6]	Constraint programming (CP)	Factory job shop with due dates and setup times; 20 instances	Energy cost −5.3%, peak −11.8%, emissions −8.3% (average)	Goals ranked by hand in a fixed order
[7]	Shifting heuristics in simulation	Delaying flexible jobs; grid data from 4 regions, 2020	Benefit depends on region and allowed delay	Grid carbon only; no on-site solar, cooling, or water
[8]	Risk-aware day-ahead optimization (ML forecasts as input)	Hourly capacity limits for flexible work; Google fleet	Deployed; cluster power −1–2% in highest-carbon hours (one example)	Fleet-level only; no cooling or water decisions
[9]	Solar- and cooling-aware scheduling algorithm	Mixed workloads with solar, free cooling, tariffs; real and synthetic traces	Energy cost up to −46% vs. earlier methods	No heat reuse or water
[10]	Rule-style scheduler (back-fill / defer) + UPS control	Data center with solar and distributed UPS batteries	Higher solar use than existing strategies	No percentage in abstract; no cooling or water
[11]	MILP	Moving jobs and stored energy between solar mini data centers	Grid electricity up to −22% vs. moving jobs alone	Small sites; no cooling or water
[12]	MILP	Off-grid data center with solar, wind, batteries, hydrogen; 72-h windows	Shows feasible supply plans	Feasibility, not cost; no cooling or water
[13]	A* search; state = (time step, battery charge)	EV charging with solar and vehicle-to-grid; 15-min steps, 24-h horizon	3–5% lower cost than particle swarm and genetic algorithms	Admissibility of h not shown; heuristic baselines only

Ref	AI Technique	Problem / Test Cases	Evaluation & Result	Limitation
[14]	Linearized integer program + heuristic	Thermal-aware work placement and cooling; model of a real data center	Cooling power −30%+ vs. a simpler baseline	No solar, heat reuse, or water
[15]	Evolutionary + local search (ML-guided)	16 distributed data centers; 3 goals; hourly	2.1× hypervolume of next-best; up to 3.7× less carbon, 1.8× less water	Learned; no optimality proof; cooling not controlled
[16]	Rule-based controller	Chiller mode switching; operating data center, 2017	+2,064 h/year of free cooling; COP +69%; 206.9 MWh less electricity	Tuned to one plant
[17]	Fuzzy rule-based control (+ small neural model)	Air dampers and cooling-unit fans; test data center	Up to 75% fan and 16% chiller power saved vs. worst case	Compared with worst case only; airflow only
[18]	MDP solved by value iteration	Turning servers on/off; Google trace; largest case 103 servers	Optimal on/off policy and its structure	State space grows fast; identical servers and job lengths
[19]	Hybrid Bayesian network	Chiller fault diagnosis; two test chillers	Better than common methods; works with missing evidence	Diagnosis only; no planning
[20]	MPC with MILP	Chillers + cold-water tank; free, hybrid, mechanical modes; Guangzhou	Yearly energy −12.19%, −4.04%, −22.15% by mode	No job scheduling, solar, or heat reuse
[21]	Stochastic MPC (scenarios)	Data center microgrid; solar, wind, batch jobs	Lower operating cost in one case study	One case; no cooling modes or water
[22]	Day-ahead optimization	Microgrid with three types of delay-tolerant jobs	Lower cost and less wasted renewable power	One case; no cooling or water
[23]	Stochastic MILP; Benders decomposition	Co-planning a data center (tasks) and an RO plant; Jubail case	Yearly cost −86.5% vs. conventional; 17% from heat reuse	Heat only preheats RO; no MD/FO, storage, or cooling choice
[25]	MPC	Campus heating from data center waste heat + storage tank	Monthly cost up to −3.2% vs. rule-based control	Heat for buildings, not water; no job scheduling
[27]	Deep reinforcement learning (ML)	Jobs + cooling airflow; trace-based simulation	Up to 15% less energy than siloed and 10% less than joint baselines	Learned policy, not traceable; no solar, heat reuse, or water
[28]	Design-space search + carbon-aware scheduling	Sizing solar, wind, batteries; production traces	Combined scenario: 100% renewable coverage carbon-optimal in five regions	Design-time only; no cooling control or water

Closest multi-resource methods, compared on the same six dimensions:

- **WaterWise [5]** — Objective: weighted carbon and water footprint. Constraints: soft delay limits. Scheduling: region and start time. Cooling: not controlled. Water: modelled, not produced. Limitation: weak baseline; no heat reuse.
- **SHIELD [15]** — Objective: cost, carbon, and water trade-offs. Constraints: no hard deadlines. Scheduling: hourly work across sites. Cooling: modelled, not controlled. Water: modelled, not produced. Limitation: learned guidance, no optimality proof.

- **DeepEE [27]** — Objective: total energy. Constraints: no overheating, balanced load. Scheduling: job placement. Cooling: airflow controlled. Water: not modelled. Limitation: learned policy, not traceable; no solar or heat reuse.

In all three, server heat is waste: none recovers it or trades cooling against water production. For CASSIA, WaterWise's soft limits would allow late high-priority jobs, and DeepEE's learned cooling choices could not be audited.

Threats to validity. Most results come from simulations, and [16], [17], [20], and [23] each cover one site, so the gains may not carry over to a large AI data center. For several newer studies we read the abstract or publisher record, not the full text (see Appendix A).

4. Research Gap and Position of CASSIA

In the studies reviewed, we see two gaps.

- **No joint plan for jobs, power, cooling, and heat-to-water at one site.** No study covers the whole chain. Haghshenas et al. [9] schedule jobs with solar and cooling in mind but reuse no heat. Zhu et al. [20] choose cooling modes with storage but schedule no jobs. Hou et al. [25] reuse heat with storage, but for buildings, not water. Yang et al. [23] link jobs to water, but heat only preheats RO in a separate plant. DeepEE [27] couples jobs and cooling airflow, and Carbon Explorer [28] is a design-time study with no cooling control or water. We did not identify an approach that schedules jobs, solar power, and cooling while routing waste heat among water production, heating, and storage.
- **No inspectable, multi-stage decision pipeline.** We call a method inspectable if every decision can be traced to a named rule, constraint, or cost. Rules [16], fuzzy control [17], CP [6], MILP [5], [23], and MPC [20], [25] meet this but each covers one or two stages; SHIELD [15] and DeepEE [27] cover more resources but rely on learned components.

Why combining existing tools is not enough. The stages form a chain: job scheduling → heat generation → cooling → heat recovery → water production. Each choice limits the next stage, so separate tools, even used together, miss these links and may conflict. CASSIA therefore needs a linked pipeline.

With one site, hourly decisions, hard deadlines, and safety-critical cooling, CASSIA (Report 1) uses exact, inspectable methods in a linked pipeline, re-planned hourly with new forecasts, as MPC does in [20] and [25]. Rules inspired by [16] pick the cooling mode (free, evaporative, or chiller), and new CASSIA rules pick the water source. A CSP then schedules jobs and cooling and assigns solar power, with deadlines, thermal limits, and zero throttling for high-priority jobs as hard constraints, and grid power and city water as weighted costs, avoiding the fixed goal ranking of [6]. A* then allocates the spare heat over 24 hours to heating loops, MD/FO water production, or thermal storage, on states (hour, storage level). As [13] shows, A* is optimal only if its heuristic never overestimates; CASSIA must prove this for its own model, and even then A* is optimal only for the heat plan given the CSP output. These are proposed design choices; Report 3 will test CASSIA against a baseline on grid electricity, municipal water, deadline compliance, and runtime.

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Appendix A — Verification Log

This appendix does not count toward the page limit. The "Checked against" column lists the source used to confirm each detail. All 28 links were opened and checked by the team.

Ref	Title (short)	Venue, year	DOI / link	Checked against	Opened
[1]	Energy and AI	IEA report, 2025	iea.org/reports/energy-and-ai	IEA executive summary	☑
[2]	Making AI less "thirsty"	Commun. ACM, 2025	doi.org/10.1145/3724499	Author PDF; dblp	☑
[3]	Data centre water consumption	npj Clean Water, 2021	doi.org/10.1038/s41545-021-00101-w	DOAJ record and abstract	☑
[4]	Carbon-aware workflow scheduling	ICPP '25, 2025	doi.org/10.1145/3754598.3754673	Full text (arXiv:2507.08725)	☑
[5]	WaterWise	PPoPP '25, 2025	doi.org/10.1145/3710848.3710891	Full text (arXiv:2501.17944)	☑
[6]	Energy-aware job shop, CP	Comput. Ind. Eng., 2025	doi.org/10.1016/j.cie.2025.111065	TU Dresden portal abstract	☑
[7]	Let's wait awhile	Middleware '21, 2021	doi.org/10.1145/3464298.3493399	Full text (arXiv:2110.13234)	☑
[8]	Carbon-aware computing for datacenters	IEEE Trans. Power Syst., 2023	doi.org/10.1109/TPWRS.2022.3173250	IEEE Xplore abstract; arXiv:2106.11750	☑
[9]	Infrastructure-aware workload scheduling	IEEE Trans. Cloud Comput., 2022	doi.org/10.1109/TCC.2020.2977040	IEEE abstract; Univ. Groningen record	☑
[10]	REDUX3 renewable-aware scheduler	IEEE NAS, 2022	doi.org/10.1109/NAS55553.2022.9925479	IEEE CSDL abstract; researchr record	☑
[11]	Data and energy migrations, mini data centers	IEEE Trans. Sustain. Comput., 2023	doi.org/10.1109/TSUSC.2022.3197090	HAL author version; dblp	☑
[12]	Stand-alone renewable data center, ILP	J. Scheduling, 2021	doi.org/10.1007/s10951-021-00700-y	HAL record; author version	☑
[13]	EV energy management, graph search	Energies, 2024	doi.org/10.3390/en17215278	Full text (MDPI)	☑
[14]	Linearized workload and cooling management	IEEE Trans. Autom. Sci. Eng., 2025	doi.org/10.1109/TASE.2024.3395471	McMaster record; arXiv:2304.04731	☑
[15]	SHIELD	IGSC '23, 2023	doi.org/10.1145/3634769.3634810	Full text (Colorado State repository)	☑
[16]	Rule-based control of chillers	Energies, 2022	doi.org/10.3390/en15072419	RePEc abstract	☑
[17]	Fuzzy airflow control	Control Eng. Pract., 2021	doi.org/10.1016/j.conengprac.2021.104833	NSF PAR author copy; British Library record	☑

Ref	Title (short)	Venue, year	DOI / link	Checked against	Opened
[18]	Discrete-time MDP policy	SMARTGREENS, 2023	doi.org/10.5220/0011846300003491	Full text (SciTePress)	☑
[19]	Hybrid Bayesian network, chillers	Eng. Appl. Artif. Intell., 2023	doi.org/10.1016/j.engappai.2023.106768	Abstract; citing record for volume	☑
[20]	MPC, hybrid cooling + cold storage	Energy, 2024	doi.org/10.1016/j.energy.2024.130304	RePEc/IDEAS record and abstract	☑
[21]	Stochastic MPC, data center microgrid	Front. Energy Res., 2022	doi.org/10.3389/fenrg.2022.863292	Open-access full text	☑
[22]	Data center microgrid, workload shift	IEEE Access, 2024	doi.org/10.1109/ACCESS.2024.3432120	DOAJ record and abstract	☑
[23]	Data center-desalination coordination	Appl. Energy, 2026	doi.org/10.1016/j.apenergy.2026.127428	Abstract page	☑
[24]	Waste heat recoveries in data centers (review)	Renew. Sustain. Energy Rev., 2023	doi.org/10.1016/j.rser.2023.113777	Open-access full text (Aalto)	☑
[25]	MPC, heat prosumer with data center waste heat	Energy, 2023	doi.org/10.1016/j.energy.2022.126579	NTNU accepted manuscript; IDEAS record	☑
[26]	Forward osmosis coupled with RO/MD	Desalination, 2022	doi.org/10.1016/j.desal.2022.116083	Open-access full text (PoliTo); DTU record	☑
[27]	DeepEE (parameterized DRL)	IEEE Trans. Serv. Comput., 2023	doi.org/10.1109/TSC.2022.3184835	BIT repository record; dblp	☑
[28]	Carbon Explorer	ASPLOS '23, 2023	doi.org/10.1145/3575693.3575754	arXiv:2201.10036; PSU/ASU records	☑

All 28 references are from 2021–2026; 7 are IEEE publications. Of the 23 reviewed studies, 21 use classical AI or optimization at their decision core; [15] and [27] are ML comparators. [1] is an agency report used only for background numbers, and [24] is a review used only for background.

Appendix B — AI Evaluation, Changes, and AI Use

(a) AI evaluation. The earlier 12-source draft was pasted into a new AI chat and evaluated with Prompt 2B. Score: **8.2 / 10** (before the changes below). This is an AI-assisted self-assessment, not an instructor grade.

Criterion	Score	Strengths	Weaknesses	Recommended fix
Relevance and quality of papers (2)	1.8	Relevant, recent ACM, IEEE, and journal papers; cross-domain choices justified	Limited coverage of directly integrated water, cooling, and scheduling work	Justify why the selection covers the problem

Criterion	Score	Strengths	Weaknesses	Recommended fix
Critical comparison (3)	2.3	Compares exactness, scalability, baselines, uncertainty, interpretability, validity	Numbers hard to compare across settings; closest methods not compared side by side	Compare WaterWise, SHIELD, and DeepEE on objectives and constraints
Research gaps (2)	1.7	Two clear gaps supported by DeepEE, WaterWise, Carbon Explorer	Gap not established beyond the reviewed literature	Keep qualified wording; state the missing integration precisely
Technical accuracy (1)	0.7	Good grasp of MILP, CP, A*, MDP, RL, heuristic search	CASSIA's heuristic not yet validated for its own cost model	Validate the heuristic against CASSIA's transition and cost model
IEEE references and organization (2)	1.7	Consistent numbering, DOIs, logical sections	Minor description details	Align descriptions with the original papers

(b) Changes made in response to the review, a Report 1 alignment check, and instructor feedback.

- Added the verified scale details of [4] (34 workflows of up to 30,000 tasks; ILP only up to about 200 tasks).
- Compared WaterWise, SHIELD, and DeepEE side by side on objective, constraints, scheduling, cooling, water, and limitation, and stated the consequences for CASSIA.
- Removed the heat-pump assumption from CASSIA's heuristic, since Report 1 specifies none; CASSIA must now prove admissibility for its own cost model.
- Aligned Section 4 with Report 1: CSP job and cooling scheduling, rule-based cooling and water selection, A* heat allocation to heating loops, MD/FO, or thermal storage, and zero thermal-throttling for high-priority jobs.
- Expanded from 12 to 28 sources, as the instructor requested: added 16 peer-reviewed sources from 2021–2026 (six from IEEE Xplore) covering carbon- and solar-aware scheduling, fuzzy control, Bayesian networks, model predictive control, and waste-heat reuse, and updated Table 1, the comparison, and the research gap to match.

(c) AI tools used. Claude (Anthropic) was used for paper discovery and summarization, drafting, and reference formatting; ChatGPT assisted with critical review, revisions, and the Prompt 2B evaluation; the team checked each paper against its source.